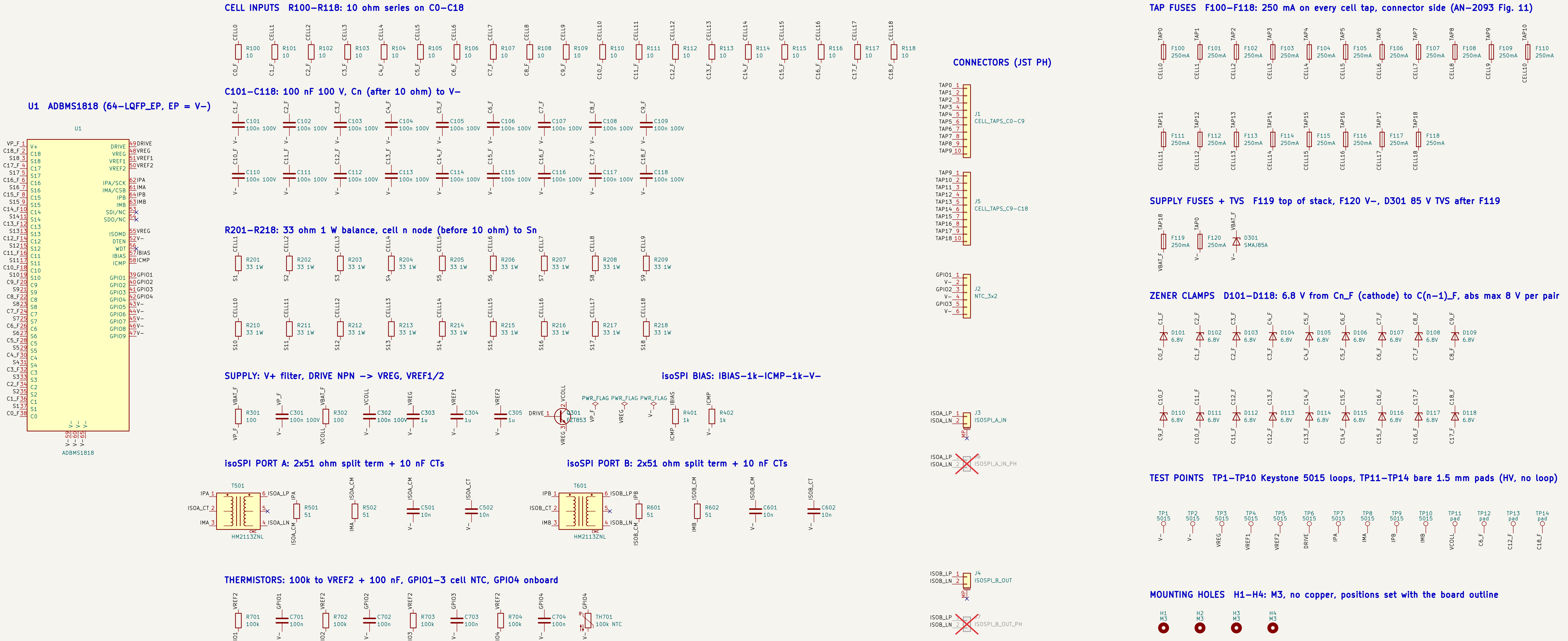
Wired with net labels: labels with the same name are one net. TAPn = connector side of fuse F1nn, CELLn = after it. V− (chip ground) has its own fuse F120.
ADBMS1818 Figure 112, isoSPI filtering per Fig. 104(B). See ../docs/slave−18s−design−basis.md.



CELL INPUTS R100–R118: 10 ohm series on C0–C18

C101–C118: 100 nF 100 V, Cn (after 10 ohm) to V−

U1 ADBMS1818 (64−LQFP_EP, EP = V−)

R201–R218: 33 ohm 1 W balance, cell n node (before 10 ohm) to Sn

SUPPLY: V+ filter, DRIVE NPN −> VREG, VREF1/2

isoSPI BIAS: IBIAS−1k−ICMP−1k−V−

isoSPI PORT A: 2x51 ohm split term + 10 nF CTs

isoSPI PORT B: 2x51 ohm split term + 10 nF CTs

THERMISTORS: 100k to VREF2 + 100 nF, GPIO1−3 cell NTC, GPIO4 onboard

CONNECTORS (JST PH)

TAP FUSES F100–F118: 250 mA on every cell tap, connector side (AN−2093 Fig. 11)

SUPPLY FUSES + TVS F119 top of stack, F120 V−, D301 85 V TVS after F119

ZENER CLAMPS D101–D118: 6.8 V from Cn_F (cathode) to C(n−1)_F, abs max 8 V per pair

TEST POINTS TP1–TP10 Keystone 5015 loops, TP11–TP14 bare 1.5 mm pads (HV, no loop)

MOUNTING HOLES H1–H4: M3, no copper, positions set with the board outline

R100 10 CELL0 C0_F
R101 10 CELL1 C1_F
R102 10 CELL2 C2_F
R103 10 CELL3 C3_F
R104 10 CELL4 C4_F
R105 10 CELL5 C5_F
R106 10 CELL6 C6_F
R107 10 CELL7 C7_F
R108 10 CELL8 C8_F
R109 10 CELL9 C9_F
R110 10 CELL10 C10_F
R111 10 CELL11 C11_F
R112 10 CELL12 C12_F
R113 10 CELL13 C13_F
R114 10 CELL14 C14_F
R115 10 CELL15 C15_F
R116 10 CELL16 C16_F
R117 10 CELL17 C17_F
R118 10 CELL18 C18_F
C101 100n 100V C1_F V−
C102 100n 100V C2_F V−
C103 100n 100V C3_F V−
C104 100n 100V C4_F V−
C105 100n 100V C5_F V−
C106 100n 100V C6_F V−
C107 100n 100V C7_F V−
C108 100n 100V C8_F V−
C109 100n 100V C9_F V−
C110 100n 100V C10_F V−
C111 100n 100V C11_F V−
C112 100n 100V C12_F V−
C113 100n 100V C13_F V−
C114 100n 100V C14_F V−
C115 100n 100V C15_F V−
C116 100n 100V C16_F V−
C117 100n 100V C17_F V−
C118 100n 100V C18_F V−
U1
VP_F 1 V+
C18_F 2 C18
S18 3 S18
C17_F 4 C17
S17 5 S17
C16_F 6 C16
S16 7 S16
C15_F 8 C15
S15 9 S15
C14_F 10 C14
S14 11 S14
C13_F 12 C13
S13 13 S13
C12_F 14 C12
S12 15 S12
C11_F 16 C11
S11 17 S11
C10_F 18 C10
S10 19 S10
C9_F 20 C9
S9 21 S9
C8_F 22 C8
S8 23 S8
C7_F 24 C7
S7 25 S7
C6_F 26 C6
S6 27 S6
C5_F 28 C5
S5 29 S5
C4_F 30 C4
S4 31 S4
C3_F 32 C3
S3 33 S3
C2_F 34 C2
S2 35 S2
C1_F 36 C1
S1 37 S1
C0_F 38 C0
DRIVE 49 DRIVE
VREG 48 VREG
VREF1 51 VREF1
VREF2 50 VREF2
IPA/SCK 62 IPA
IMA/CSB 61 IMA
IPB 64 IPB
IMB 63 IMB
SDI/NC 53
SDO/NC 54
ISOMD 55 VREG
DTEN 52 V−
WDT 56
IBIAS 57 IBIAS
ICMP 58 ICMP
GPIO1 39 GPIO1
GPIO2 40 GPIO2
GPIO3 41 GPIO3
GPIO4 42 GPIO4
GPIO5 43 V−
GPIO6 44 V−
GPIO7 45 V−
GPIO8 46 V−
GPIO9 47 V−
V− 59 V−
V− 60 V−
V− 65 V−
ADBMS1818
R201 33 1W CELL1 S1
R202 33 1W CELL2 S2
R203 33 1W CELL3 S3
R204 33 1W CELL4 S4
R205 33 1W CELL5 S5
R206 33 1W CELL6 S6
R207 33 1W CELL7 S7
R208 33 1W CELL8 S8
R209 33 1W CELL9 S9
R210 33 1W CELL10 S10
R211 33 1W CELL11 S11
R212 33 1W CELL12 S12
R213 33 1W CELL13 S13
R214 33 1W CELL14 S14
R215 33 1W CELL15 S15
R216 33 1W CELL16 S16
R217 33 1W CELL17 S17
R218 33 1W CELL18 S18
R301 100 VBAT_F VP_F
C301 100n 100V VP_F V−
R302 100 VBAT_F VCOLL
C302 100n 100V VCOLL V−
C303 1u VREG V−
C304 1u VREF1 V−
C305 1u VREF2 V−
DRIVE 1 Q301 FZT853 2 VCOLL 3 VREG
PWR_FLAG VP_F
PWR_FLAG VREG
PWR_FLAG V−
R401 1k IBIAS ICMP
R402 1k ICMP V−
T501 HM2113ZNL CMC
IPA 1 ISOA_CT 2 IMA 3 6 ISOA_LP 5 4 ISOA_LN
R501 51 IPA ISOA_CM
R502 51 ISOA_CM IMA
C501 10n ISOA_CM V−
C502 10n ISOA_CT V−
T601 HM2113ZNL CMC
IPB 1 ISOB_CT 2 IMB 3 6 ISOB_LP 5 4 ISOB_LN
R601 51 IPB ISOB_CM
R602 51 ISOB_CM IMB
C601 10n ISOB_CM V−
C602 10n ISOB_CT V−
R701 100k VREF2 GPIO1
C701 100n GPIO1 V−
R702 100k VREF2 GPIO2
C702 100n GPIO2 V−
R703 100k VREF2 GPIO3
C703 100n GPIO3 V−
R704 100k VREF2 GPIO4
C704 100n GPIO4 V−
TH701 100k NTC GPIO4 V−
J1 CELL_TAPS_C0−C9
TAP0 1 TAP1 2 TAP2 3 TAP3 4 TAP4 5 TAP5 6 TAP6 7 TAP7 8 TAP8 9 TAP9 10
J5 CELL_TAPS_C9−C18
TAP9 1 TAP10 2 TAP11 3 TAP12 4 TAP13 5 TAP14 6 TAP15 7 TAP16 8 TAP17 9 TAP18 10
J2 NTC_3x2
GPIO1 1 V− 2 GPIO2 3 V− 4 GPIO3 5 V− 6
J3 ISOSPI_A_IN ISOA_LP 1 ISOA_LN 2 MP
J6 ISOSPI_A_IN_PH ISOA_LP 1 ISOA_LN 2
J4 ISOSPI_B_OUT ISOB_LP 1 ISOB_LN 2 MP
J7 ISOSPI_B_OUT_PH ISOB_LP 1 ISOB_LN 2
F100 250mA TAP0 CELL0
F101 250mA TAP1 CELL1
F102 250mA TAP2 CELL2
F103 250mA TAP3 CELL3
F104 250mA TAP4 CELL4
F105 250mA TAP5 CELL5
F106 250mA TAP6 CELL6
F107 250mA TAP7 CELL7
F108 250mA TAP8 CELL8
F109 250mA TAP9 CELL9
F110 250mA TAP10 CELL10
F111 250mA TAP11 CELL11
F112 250mA TAP12 CELL12
F113 250mA TAP13 CELL13
F114 250mA TAP14 CELL14
F115 250mA TAP15 CELL15
F116 250mA TAP16 CELL16
F117 250mA TAP17 CELL17
F118 250mA TAP18 CELL18
F119 250mA TAP18 VBAT_F
F120 250mA TAP0 V−
D301 SMAJ85A VBAT_F V−
D101 6.8V C1_F C0_F
D102 6.8V C2_F C1_F
D103 6.8V C3_F C2_F
D104 6.8V C4_F C3_F
D105 6.8V C5_F C4_F
D106 6.8V C6_F C5_F
D107 6.8V C7_F C6_F
D108 6.8V C8_F C7_F
D109 6.8V C9_F C8_F
D110 6.8V C10_F C9_F
D111 6.8V C11_F C10_F
D112 6.8V C12_F C11_F
D113 6.8V C13_F C12_F
D114 6.8V C14_F C13_F
D115 6.8V C15_F C14_F
D116 6.8V C16_F C15_F
D117 6.8V C17_F C16_F
D118 6.8V C18_F C17_F
TP1 5015 V−
TP2 5015 V−
TP3 5015 VREG
TP4 5015 VREF1
TP5 5015 VREF2
TP6 5015 DRIVE
TP7 5015 IPA
TP8 5015 IMA
TP9 5015 IPB
TP10 5015 IMB
TP11 pad VCOLL
TP12 pad C6_F
TP13 pad C12_F
TP14 pad C18_F
H1 M3
H2 M3
H3 M3
H4 M3

Not for manufacture
Baseline: ADBMS1818 datasheet Figure 112 typical application

Sheet: /
File: adbms1818−slave−18s.kicad_sch

Title: ADBMS1818 18S slave module

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